ECA5403-ELECTROMAGNETIC FILEDS AND WAVES

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C01-1

EXPERIMENT 1

AIM:

To write a MATLAB program for transformation between Cartesian, Cylindrical and Spherical coordinate systems.

PROGRAM:

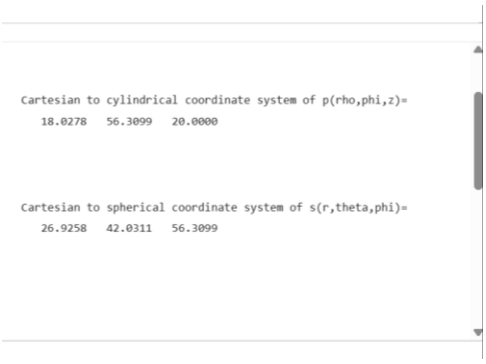
```
clc;
x=input('Enter the value of X=');
y=input('Enter the value of y=');
z=input('Enter the value of z=');
% Cartesian to Cylindrical
rho=sqrt(x^2+y^2);
phi=atan2d(y,x);
disp('Cartesian to cylindrical coordinate system of p(rho,phi,z)=');
disp([rho phi z]);
% Cartesian to Spherical
r=sqrt(x^2+y^2+z^2);
theta=acosd(z/r);
phi=atan2d(y,x);
disp('Cartesian to spherical coordinate system of s(r,theta,phi)=');
disp([r theta phi]);
```

```
clc;
rho=input('Enter the value of rho=');
phi=input('Enter the value of phi=');
z=input('Enter the value of z=');
```

```
% Cylindrical to Cartesian
x=rho*cosd(phi);
y=rho*sind(phi);
disp('Cylindrical to Cartesian coordinate system of p(x,y,z)=');
disp([x y z]);
% Cylindrical to Spherical
r=sqrt(rho^2+z^2);
theta=acosd(z/r);
disp('Cylindrical to spherical coordinate system of s(r,theta,phi)=');
disp([r theta phi]);

clc;
r=input('Enter the value of r=');
theta=input('Enter the value of theta (deg)=');
phi=input('Enter the value of phi (deg)=');
% Spherical to Cartesian
x=r*sind(theta)*cosd(phi);
y=r*sind(theta)*sind(phi);
z=r*cosd(theta);
disp('Spherical to Cartesian coordinate system of p(x,y,z)=');
disp([x y z]);
% Spherical to Cylindrical
rho=r*sind(theta);
disp('Spherical to cylindrical coordinate system of s(rho,phi,z)=');
disp([rho phi z]);
```

OUTPUT:




```
Cylindrical to Cartesian coordinate system of p(x,y,z)=
    9.6593    2.5882   20.0000


Cylindrical to spherical coordinate system of s(r,theta,phi)=
   22.3607   26.5651   15.0000
```

```
Spherical to Cartesian coordinate system of p(x,y,z)=
    2.4321    0.8852    9.6593


Spherical to cylindrical coordinate system of s(rho,phi,z)=
    2.5882   20.0000    9.6593
```

RESULTS:

Thus the transformation between Cartesian, Cylindrical and Spherical coordinate systems have been done using MATLAB software.

S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Apply coordinate-system definitions and angle conventions	Apply coordinate-system definitions, symbols, units, and angle conventions	Applies coordinate-system definitions and angle conventions throughout.	Uses correct definitions with one minor notation or convention error.	Shows partial understanding; multiple corrections are needed.	Uses incorrect coordinate definitions or angle conventions.	25	2.5	
2	Implement transformations in MATLAB	Implement six transformations in MATLAB	Implements all six transformations correctly using suitable degree-based functions.	Implements most transformations correctly with minor coding errors.	Implements some transformations but needs assistance to correct errors.	Unable to implement the required transformations.	25	2.5	
3	Verify numerical values and reverse transformations	Verify numerical values and reverse transformations	All outputs agree with analytical values; reverse transformations recover the original point within rounding precision.	Most values agree; small unexplained differences occur.	Several values differ or reverse verification is incomplete.	Outputs are incorrect or not verified.	30	3	
4	Interpret and document results	Interpret and document results	Presents complete tables, units, deviations, and a clear conclusion.	Documentation is clear with minor omissions.	Tables or interpretation are incomplete.	Results and conclusion are missing or unclear.	20	2	
Total							100	10	